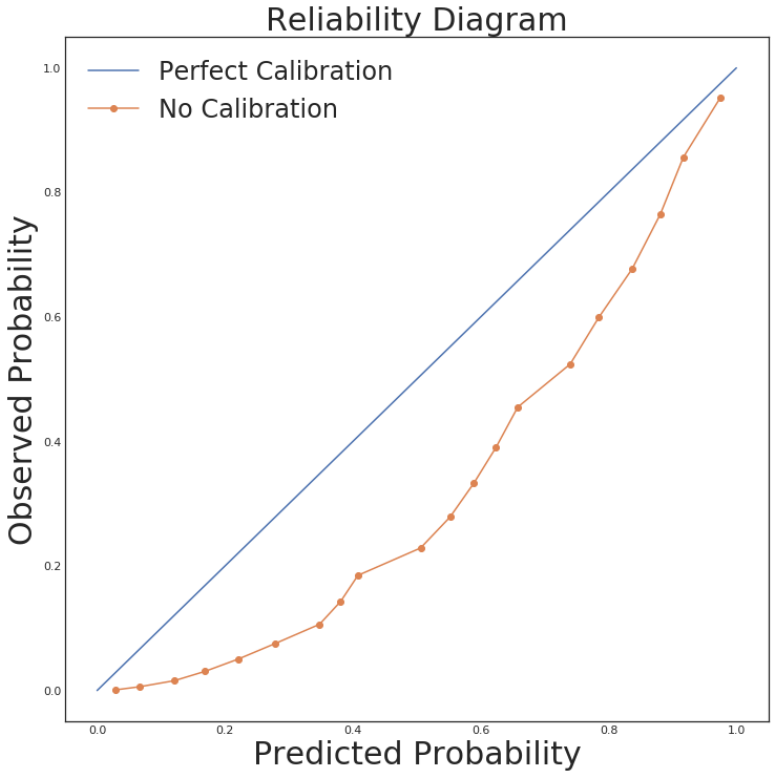
Calibration versus Discrimination



Learning goals

- Understand difference between calibration and discrimination
- How to diagnose calibration
- How to calibrate probabilities

PERFORMANCE METRICS AND USE CASE NEEDS

- **Match metric to task:** Optimize what reflects success in your application
- **Robustness to outliers:** Use *MAE* (instead of *MSE*) when occasional large errors should not dominate the metric; use *MSE* if they should
- **Class imbalance:** Use F1-score instead of accuracy
- **Unequal error costs:** Apply cost-weighted metrics for asymmetric penalties (i.e., weighting FP and FN differently)
- **Prediction type matters:**
  - *Discrete predictions:* Use confusion-matrix metrics (e.g., *F1 score*) $\rightarrow$ Focus on **discrimination** (how well classes are separated)
  - *Probabilistic predictions:* Use proper scoring rules (e.g., *Brier score*) $\rightarrow$ Consider **calibration** (matching predicted to actual probabilities)

DISCRIMINATION

Consider the true positive $Y = 1$, true negative $Y = 0$, and predicted label $\hat{Y}$.

Discrimination: Ability of $\hat{Y}$ to well-separate positive/negative instances.
$\Rightarrow$ Confusion matrix-based measures are discrimination measures, e.g., $FPR = P(\hat{Y} = 1 | Y = 0)$, $TPR = P(\hat{Y} = 1 | Y = 1)$, AUC.

Predicted Values

True Values	Positive	Negative	
Positive	TP	FN	$Sensitifity = \frac{TP}{TP + FN}$
Negative	FP	TN	$Specificity = \frac{TN}{TN + FP}$
	$Precision = \frac{TP}{TP + FP}$	$Negative\ Predictive\ Value = \frac{TN}{TN + FN}$	$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$

CALIBRATION

Consider the true positive $Y = 1$, true negative $Y = 0$, and predicted label $\hat{Y}$.

Calibration: When the predicted probabilities $\hat{p}$ closely agree with the observed proportion for $Y = 1$ (for any reasonable grouping).

- **Calibration in the large**: Observed vs. predicted prob. in *full sample*.

Predicted prob. of survival (1 = survived, 0 = not) for 6 Titanic passengers of different gender

	Probability	Actual
	0.2	0
	0.8	0
	0.9	1
	0.1	0
	0.7	1
	0.3	1

Calibration in the large:
property of full sample

Mean predicted probability of survival:

0.5

Observed probability of survival in sample:

0.5

Calibration in the large:

0.5 = 0.5 ✅

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- **Calibration in the small**: Observed vs. predicted prob. in *subsets*.

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property of full sample

Mean predicted probability of survival:

0.5

Observed probability of survival in sample:

0.5

Calibration in the large:

0.5 = 0.5 ✓

Calibration in the small:
property of subsets

Mean predicted probability of survival:

male: 0.20
female: 0.80

Observed probability of survival in sample:

male: 0.33
female: 0.67

Calibration in the small:

male: 0.20 ≠ 0.33 ✗
female: 0.80 ≠ 0.67 ✗

CALIBRATION AND DISCRIMINATION

A well-calibrated classifier can be poorly discriminating, e.g.

Obs. Nr.	true Y	$\hat{f}_1$	$\hat{f}_2$
1	1	1	0
2	1	1	0
3	0	0	1
4	0	0	1
Avg Prob	50%	50%	50%

- Both models ($\hat{f}_1$ and $\hat{f}_2$) result in the same calibration in the large (50%).
- However, $\hat{f}_1$ is better than $\hat{f}_2$ as it correctly classifies the real outcome Y.

CALIBRATION AND DISCRIMINATION

A well discriminating classifier can have a bad calibration, e.g.

Obs. Nr.	truth Y	$\hat{f}_1$	$\hat{f}_2$
1	1	0.9	0.9
2	1	0.9	0.9
3	0	0.1	0.7
4	0	0.1	0.7
Avg Prob	50%	50%	80%

- Both models are well discriminating, i.e., setting thresholds $c_1 = 0.5$ for $\hat{f}_1$ and $c_2 = 0.8$ for $\hat{f}_2$ perfectly separates positive and negative observations (and will result, e.g., in the same AUC).
- $\hat{f}_2$ is poorly calibrated as the averaged probabilities are 80% and do not match the truth proportion of positive observations (which is 50%).

$\Rightarrow$ **How can we measure calibration quality?**

CALIBRATION PLOT / RELIABILITY DIAGRAM

To assess calibration visually, we can plot on the

- x-axis: average predicted probability (e.g., grouped by quantiles)
- y-axis: observed proportion of positive instances in each group

$\hat{p}$	y
0.1	0
0.1	0
0.4	0
0.4	1
0.7	0
0.7	1
0.7	1
0.9	1

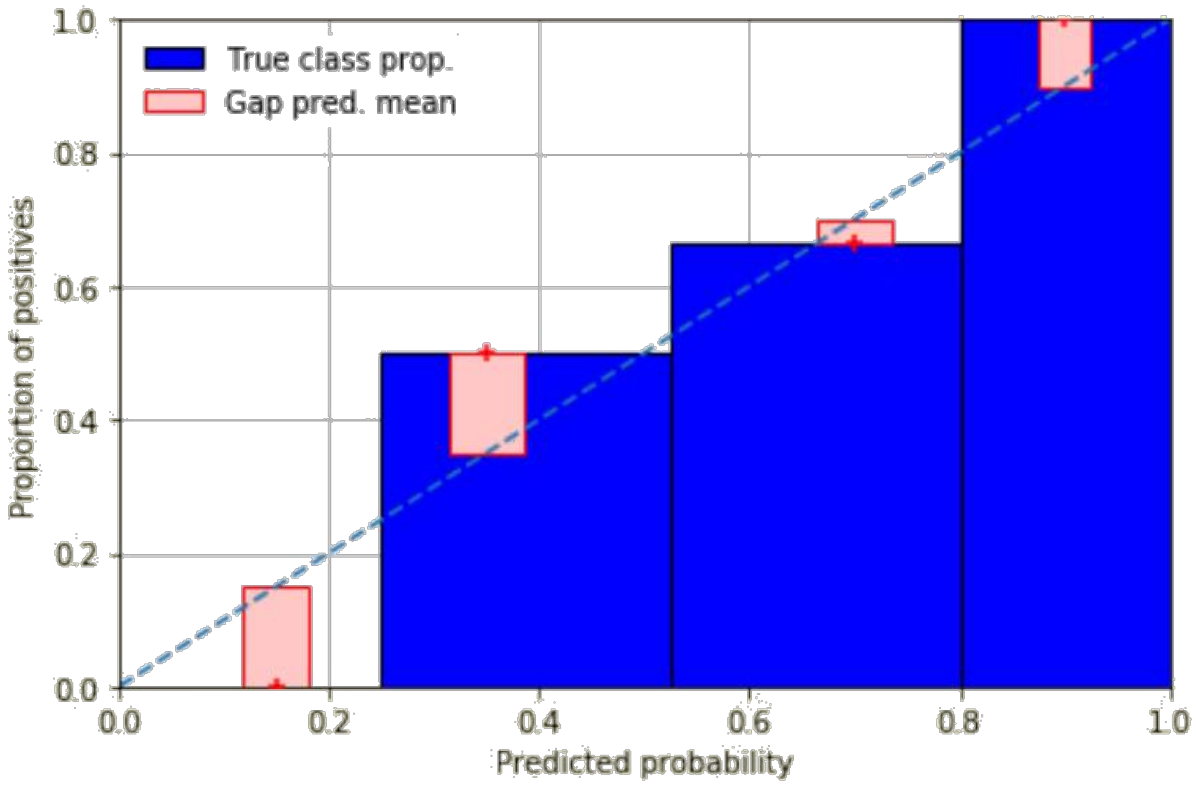
Gap pred. mean: Shows deviation between observed proportion of positives (red points) and average predicted probability within each group (red bar).

CALIBRATION PLOT / RELIABILITY DIAGRAM

Changing predictions will change the position of the red points on the x-axis

- The closer the red points to the diagonal line, the better calibration
- **Question:** Did we improve or worsen calibration?

$\hat{p}$	y
0.1	0
0.2	0
0.3	0
0.4	1
0.6	0
0.7	1
0.8	1
0.9	1

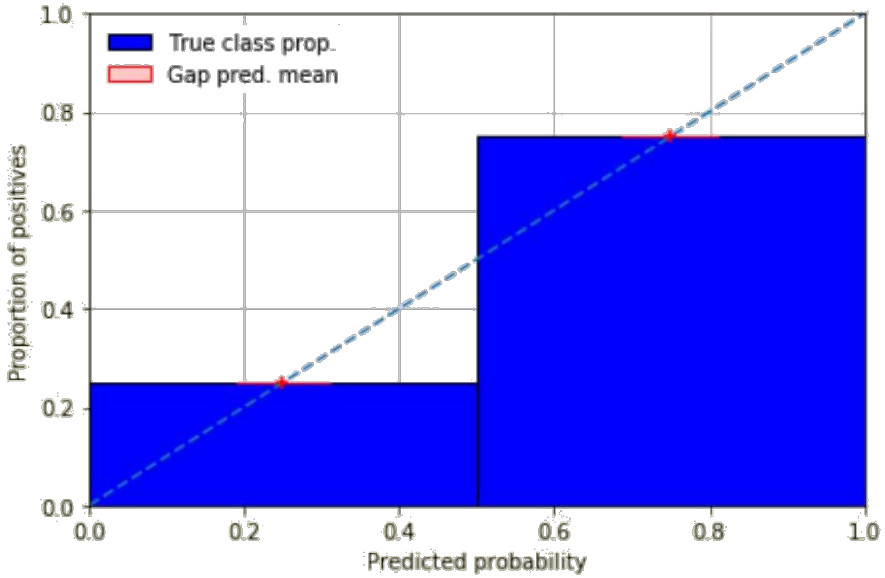


CALIBRATION PLOT / RELIABILITY DIAGRAM

Different groupings lead to a completely different picture

2 Groups

$\hat{p}$	y
0.1	0
0.2	0
0.3	0
0.4	1
0.6	0
0.7	1
0.8	1
0.9	1



CALIBRATION PLOT / RELIABILITY DIAGRAM

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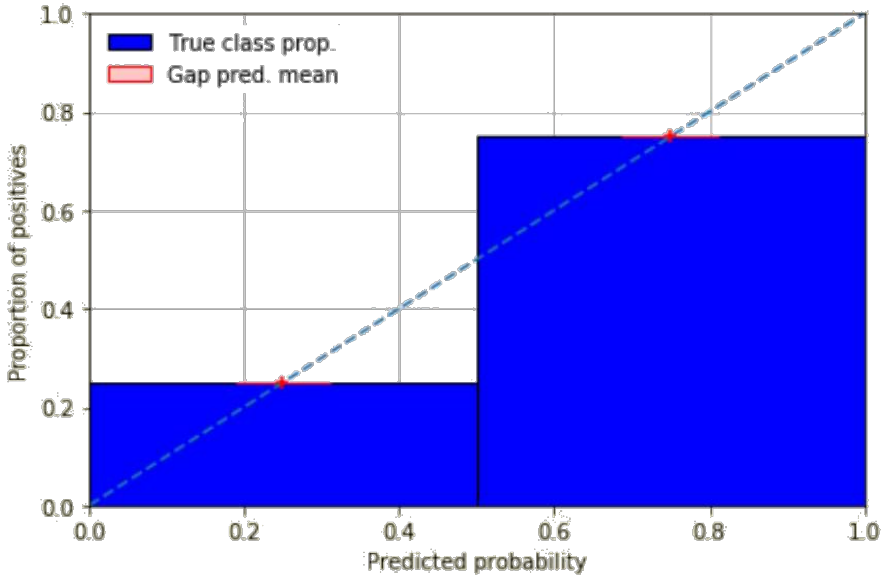
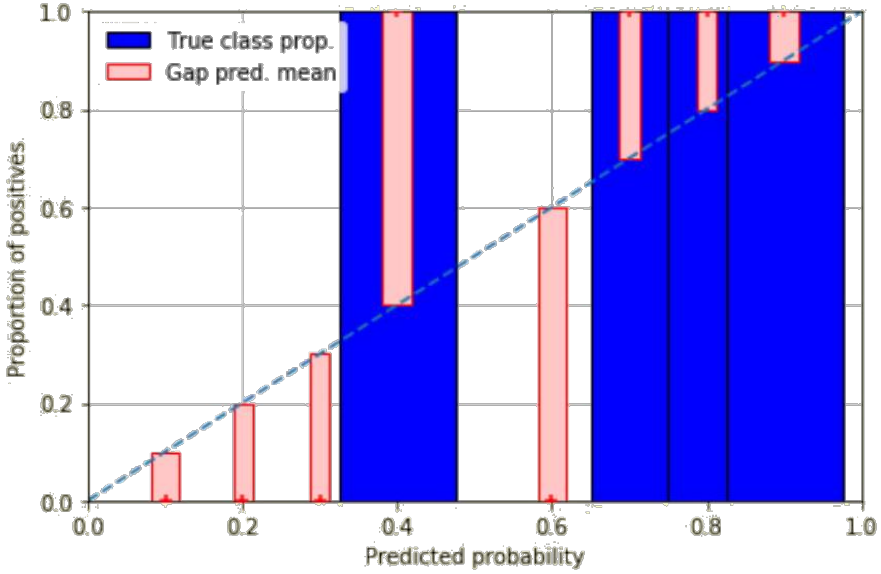


8 Groups

$\hat{p}$	y
0.1	0
0.2	0
0.3	0
0.4	1
0.6	0
0.7	1
0.8	1
0.9	1



IMPORTANCE OF GROUPING PREDICTIONS

Evaluating calibration with bins/groups:

- We need enough predictions in each group for reliable estimates.
- Trade-off:
  - large groups give better empirical estimates
  - small groups allow a more fine-grained assessment of calibration

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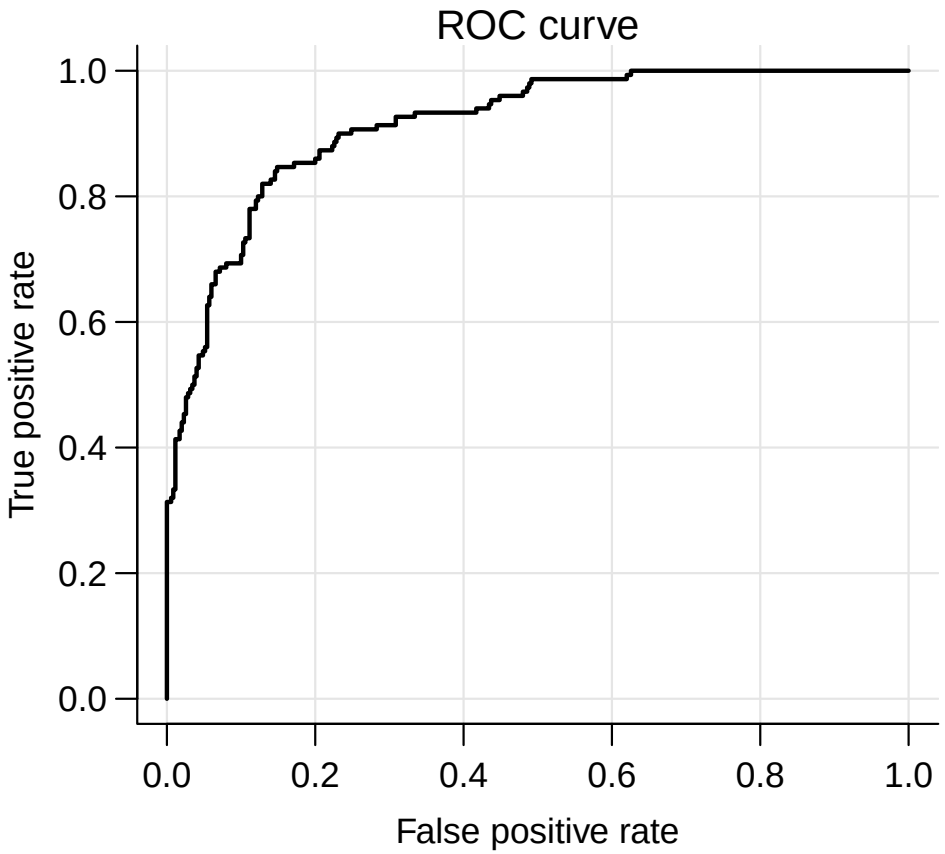
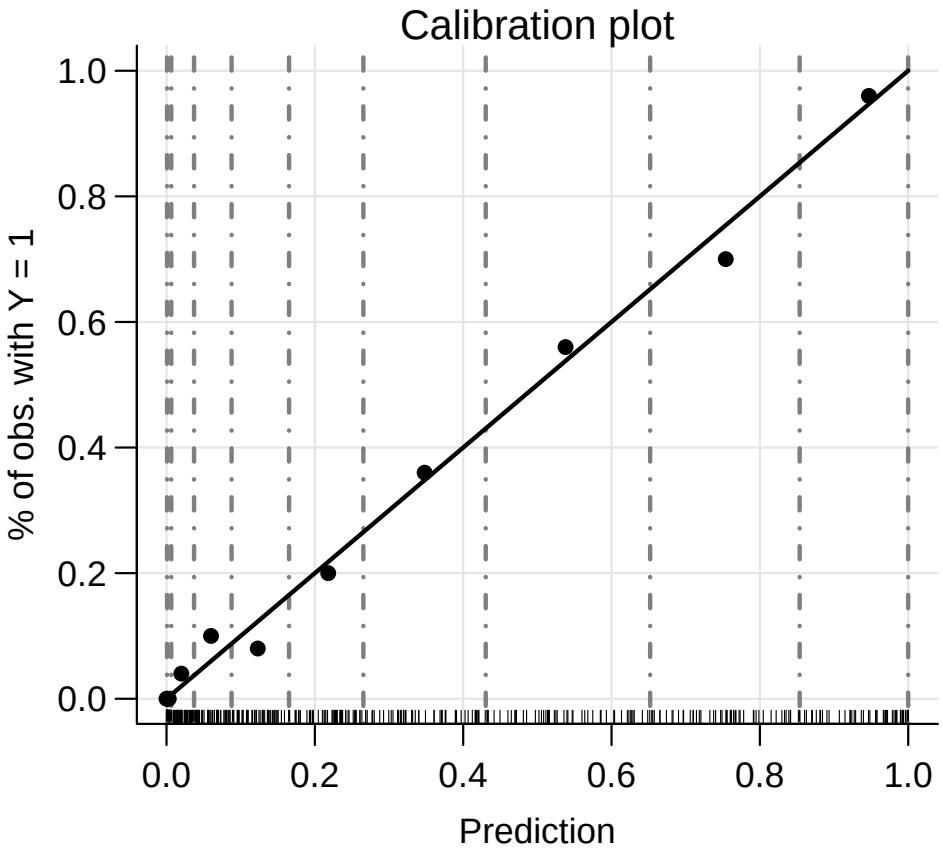
How groups/bins can be created:

- Equal-Width: Splits predicted probabilities into equally sized intervals.
  $\Rightarrow$ Can lead to bins with few or no observations.
- Equal-Frequency: Each bin holds a similar number of observations.
  $\Rightarrow$ Bin width can vary notably.

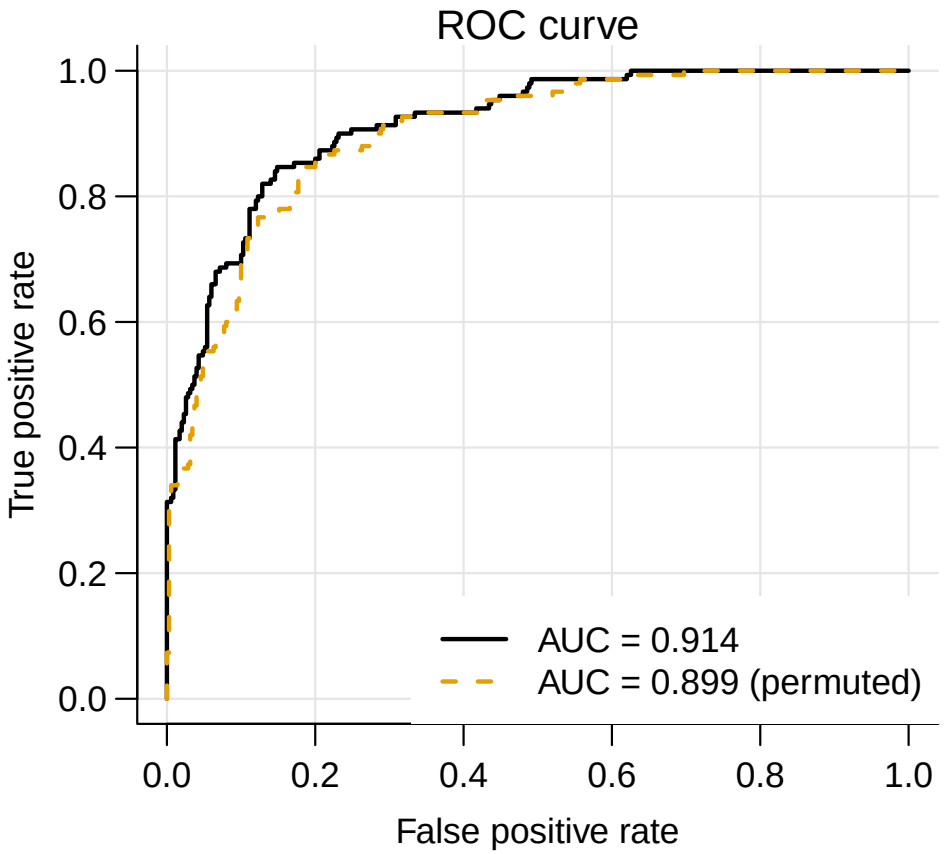
ROC CURVE VS. CALIBRATION PLOT

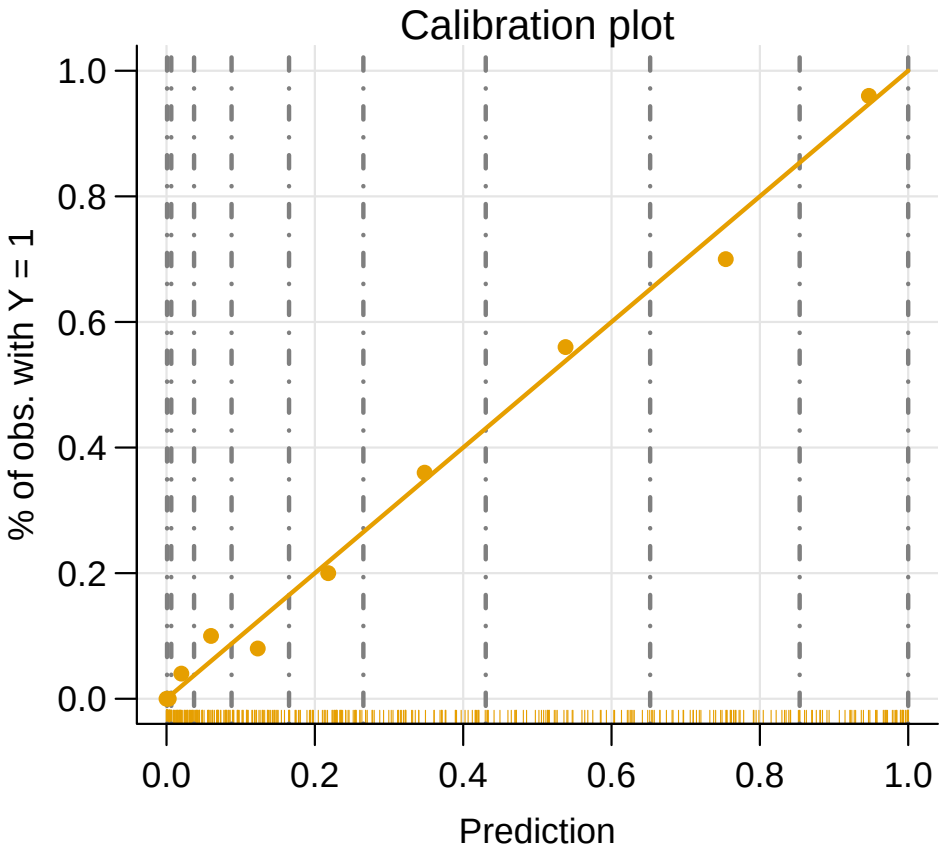




- **ROC:** Measures only discrimination as it is based on TPR and FPR and assesses only the ranking of predicted probabilities (not their magnitude).
- **Calibration plot:** Measures (for reasonable groups, here by deciles) how well the predicted probabilities match the proportion of positives.

ROC CURVE VS. CALIBRATION PLOT

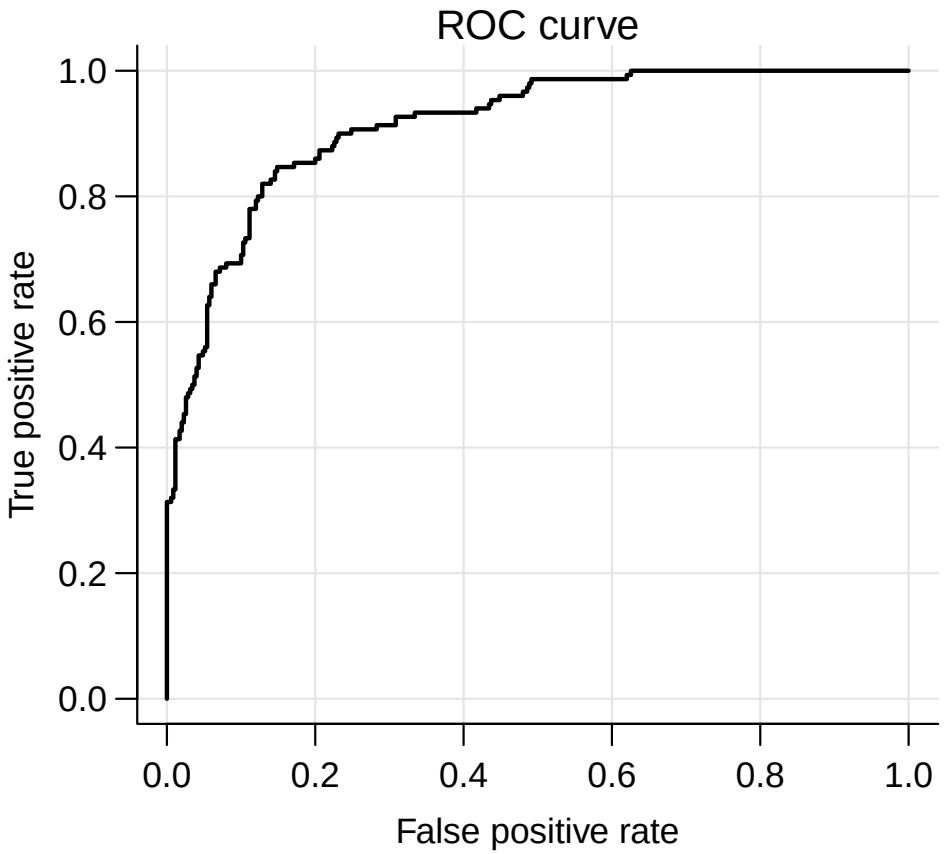


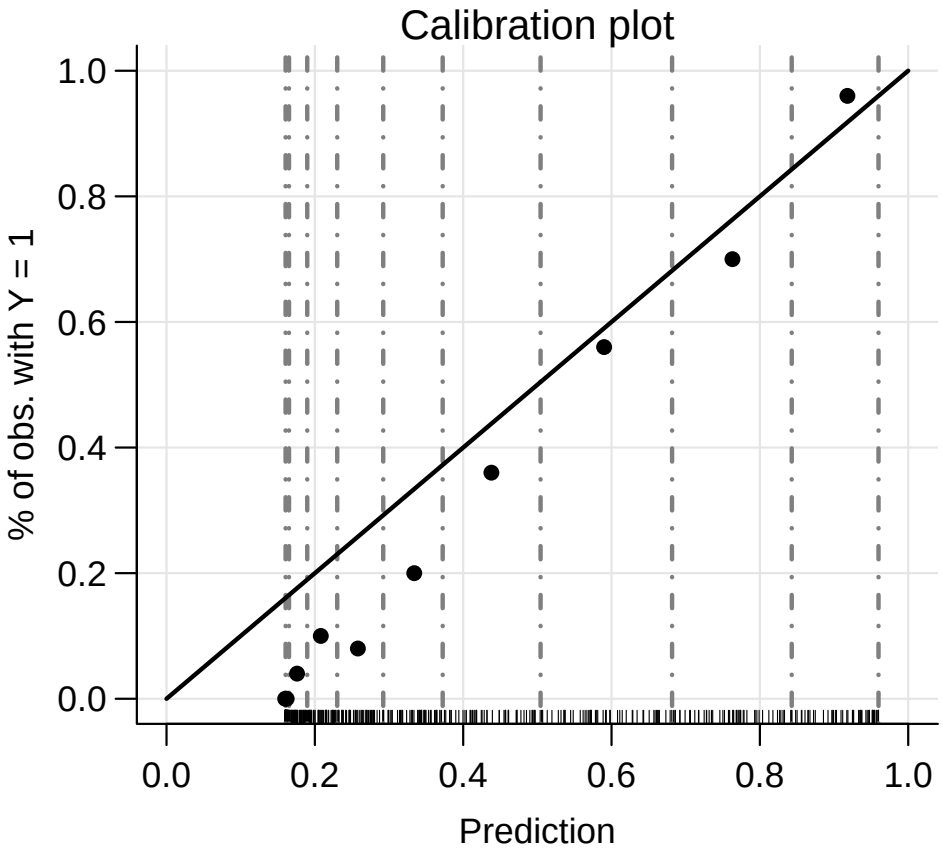


Permuting predictions within each group (e.g., randomly assigning different predictions to observations within each decile)

- worsens the ROC curve / AUC (ranking within each decile changes).
- does not affect the calibration plot (as we only look at averages).

ROC CURVE VS. CALIBRATION PLOT

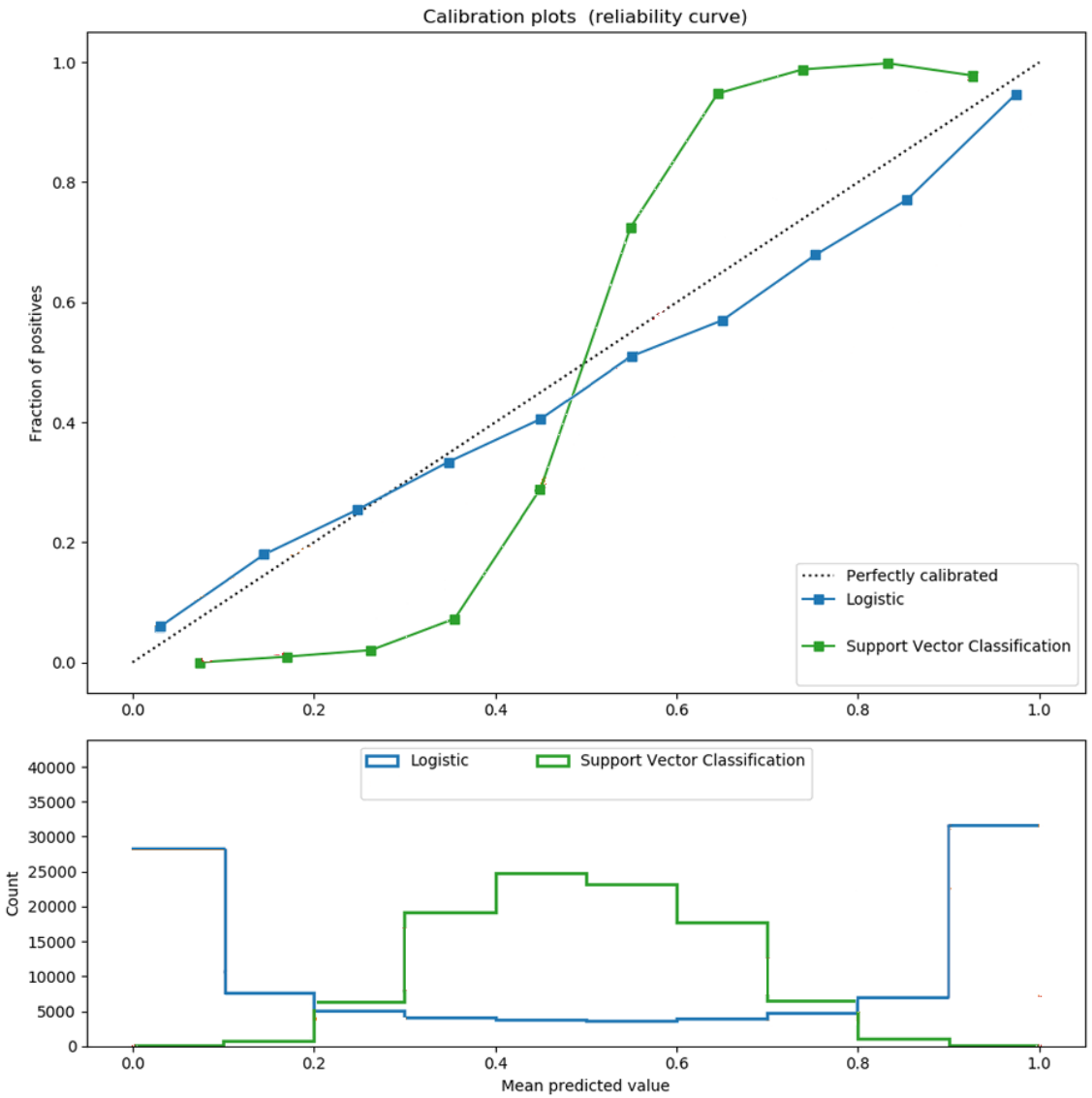




Monotonic transformations of the predicted probabilities (e.g., $\frac{\hat{p}+0.2}{1.2}$)

- do not affect the ROC curve as the ranking of $\hat{p}$ will not change.
- affect the calibration plot (here it looks worse).

EXAMPLE: CALIBRATION PLOTS (EQUAL-WIDTH)



- Many algorithms (except logistic regression) return biased probabilities.
- Linear SVC: badly calibrated, as predictions refer to a distance (to decision boundary) and not to probabilities.

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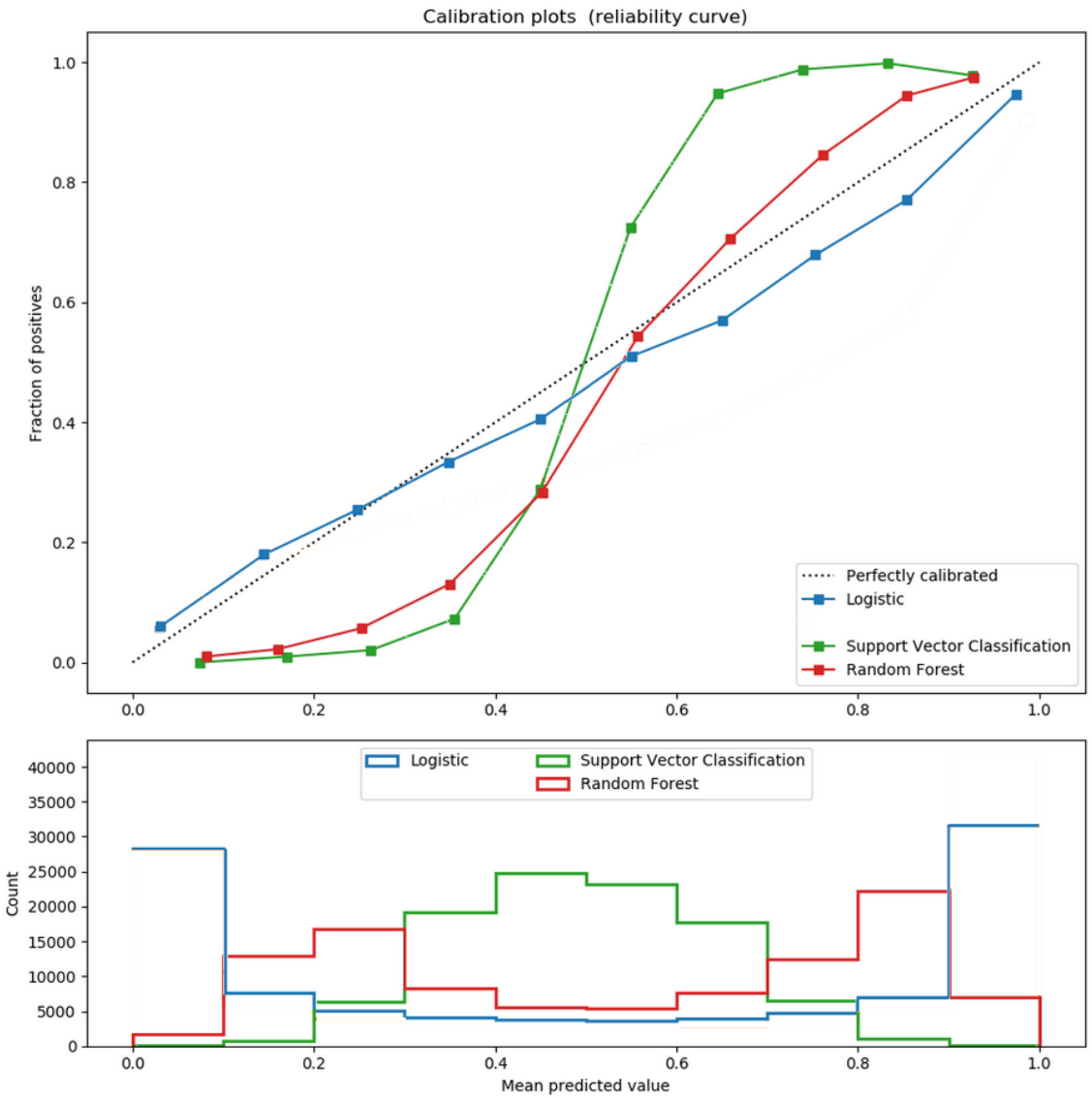

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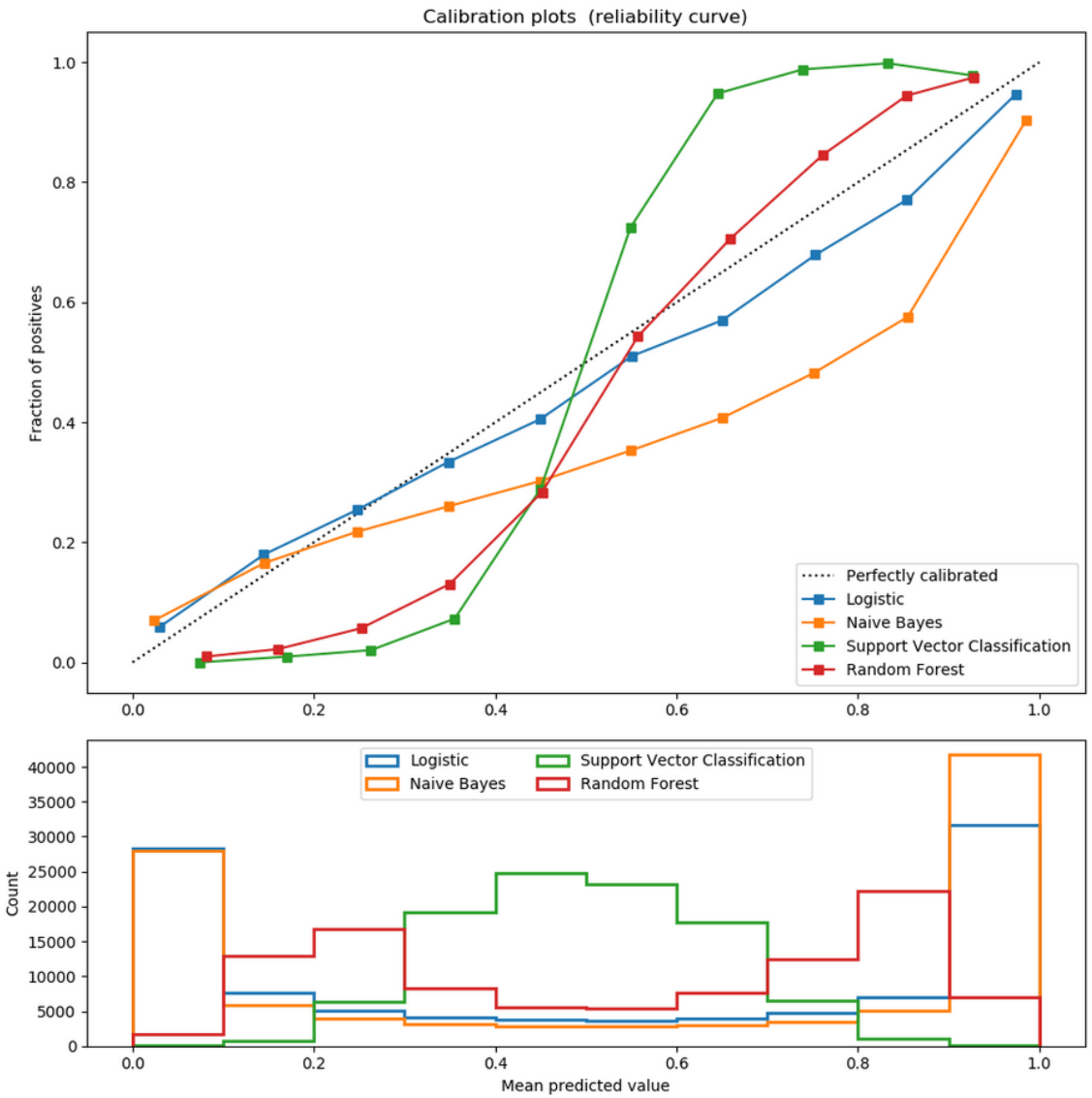


- Many algorithms (except logistic regression) return biased probabilities.
- Linear SVC: badly calibrated, as predictions refer to a distance (to decision boundary) and not to probabilities.
- Random Forest: probabilities close to 0 or 1 are very rare.
- Naive Bayes: pushes probabilities to 0 or 1 (see histograms).

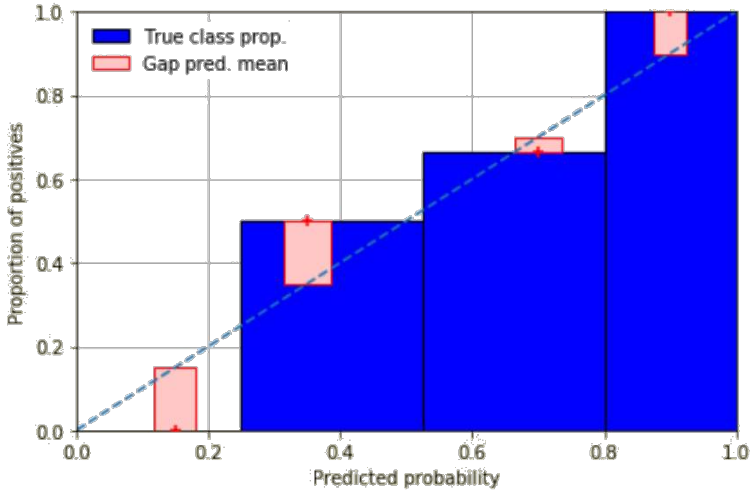
METRICS FOR CALIBRATION

Goal: Measure agreement between pred. probabilities and observed freq.

- N instances; $\hat{p} \in [0, 1]$ predicted probabilities; $y \in \{0, 1\}$ true labels
- Predictions are grouped into M disjoint bins $B_1, \ldots, B_M$
- $\bar{p}_m$: average predicted probability in bin B_m
- $\bar{y}_m$: average true label in bin B_m

Calibration-only Metrics: Measure how well predicted probabilities match observed frequencies, independent of class separation.

- **Expected Calibration Error (ECE)**: Average bin-wise gap between predicted probabilities and actual frequencies $\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} |\bar{y}_m - \bar{p}_m|$
- **Maximum Calibration Error (MCE)**: Largest bin-wise gap between predicted probabilities and actual frequencies $\mathrm{MCE} = \max_{1 \leq m \leq M} |\bar{y}_m - \bar{p}_m|$



HYPOTHESIS TEST FOR CALIBRATION

- **Goal**: Test if a classifier is really uncalibrated on the given data set
- Group instances into M bins based on percentiles of the predicted probabilities for the positive class (equal-frequency binning)
- Compute the test statistics of the Hosmer-Lemeshow (HL) test

$$H = \sum_{m=1}^{M} \left(\frac{(O_m^+ - E_m^+)^2}{E_m^+} + \frac{(O_m^- - E_m^-)^2}{E_m^-} \right)$$

  - M number of bins
  - O_m^+ observed positives or O_m^- observed negatives in bin m
  - $E_m^+ = N_m \times \bar{p}_m$ expected positives
  - $E_m^- = N_m \times (1 - \bar{p}_m)$ expected negatives in bin m
  - N_m number of obs. and $\bar{p}_m$ average predictions in bin m
- If H is small $\Rightarrow$ observed outcomes match predicted probabilities (well-calibrated model)
- Follows a Chi-Squared distribution with $M - 2$ degrees of freedom

HL TEST: REVISITED EXAMPLE

Obs. Nr.	truth Y	$\hat{f}_1$	$\hat{f}_2$
1	1	0.9	0.9
2	1	0.9	0.9
3	0	0.1	0.7
4	0	0.1	0.7

Group obs. by their predicted probability, creating $M = 2$ bins for each model.

Model $\hat{f}_1$ (Well-calibrated)

Bin ($\hat{p}$)	O^+	E^+	O^-	E^-
0.9	2	1.8	0	0.2
0.1	0	0.2	2	1.8

$$H = \left(\frac{(0-0.2)^2}{0.2} + \frac{(2-1.8)^2}{1.8}\right) + \left(\frac{(2-1.8)^2}{1.8} + \frac{(0-0.2)^2}{0.2}\right) = 0.44$$

Model $\hat{f}_2$ (Poorly calibrated)

Bin ($\hat{p}$)	O^+	E^+	O^-	E^-
0.9	2	1.8	0	0.2
0.7	0	1.4	2	0.6

$$H = \left(\frac{(0-1.4)^2}{1.4} + \frac{(2-0.6)^2}{0.6}\right) + \left(\frac{(2-1.8)^2}{1.8} + \frac{(0-0.2)^2}{0.2}\right) = 4.89$$

Conclusion: H-L statistic for $\hat{f}_2$ is over 10 times larger due to a large discrepancy for the $\hat{p} = 0.7$ bin ($O^+ = 0$ vs. $E^+ = 1.4$).

CALIBRATION: OVER- AND UNDER-ESTIMATES

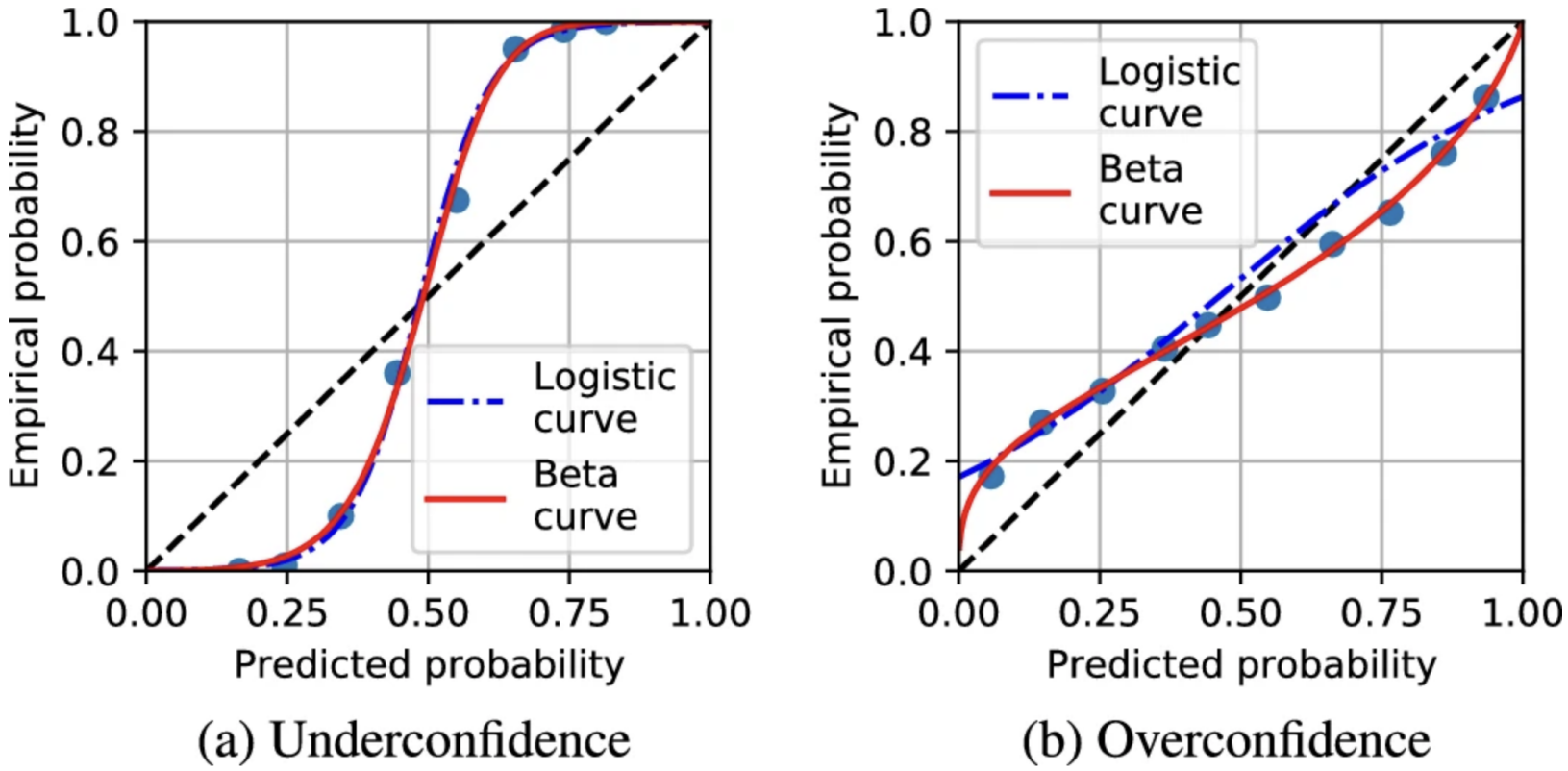
Example: Weather model predicting daily rain probabilities

- Calibration implies: predicted probabilities should (on average) match their observed frequencies
  $\Rightarrow$ E.g., '70% chance of rain' $\rightarrow$ should rain $\approx 70\%$ of the time
  $\Rightarrow$ Ensures predictions can be interpreted as actual risk/probabilities
- Consider the following predictions of a (good performing) classifier:

$\hat{p}$	y
0.1	0
0.1	0
0.4	0
0.4	1
0.7	0
0.7	1
0.7	1
0.9	1

- '10% chance of rain' was a slight over-estimate ($\bar{y} = 0/2 = 0\%$).
- '40% chance of rain' was a slight under-estimate ($\bar{y} = 1/2 = 50\%$).
- '70% chance of rain' was a slight over-estimate ($\bar{y} = 2/3 = 67\%$).
- '90% chance of rain' was a slight under-estimate ($\bar{y} = 1/1 = 100\%$).

CALIBRATION: OVER- AND UNDER-ESTIMATES



Filho et al. (2023)

- **Underconfidence**: Predicted probabilities are too close to 0.5 and need to be expanded (classifier is generally under-estimating)
- **Overconfidence**: Predicted probabilities are too close to 0 or 1 and need to be compressed towards the middle (classifier is over-estimating)

CALIBRATING PROBABILITIES

Goal: Calibration Techniques

- Calibrating probabilities involves **post-processing** predictions.
- **Aim:** Ensure predicted probabilities of any event should (on average) match their observed empirical probabilities.
  $\Rightarrow$ Makes predictions interpretable as actual risk/probabilities.
- Calibration should be performed on **new data** not used for model fitting.
- Link function in **logistic regression** can be viewed as a calibration of predictions of a **linear regression**.

CALIBRATING PROBABILITIES

- Let $s(\mathbf{x})$ denote the predicted (uncalibrated) score for input $\mathbf{x}$
- Define $\mathbb{S}$ as the set of possible scores produced by the classifier
- Goal: Construct a *calibration function* C that maps scores $s(\mathbf{x}) \in \mathbb{S}$ to calibrated probabilities $C(s(\mathbf{x})) \in [0, 1]$:

$$C : \mathbb{S} \to [0, 1], \text{such that } C(s(\mathbf{x})) \approx P(y = 1 \mid s(\mathbf{x})) \text{ (well-calibrated)}$$

- To learn calibrator function C, use a separate *calibration dataset*:

$$\mathcal{D}_{\text{cal}} = \{(s^{(i)}, y^{(i)})\}_{i=1}^{n} \subset \mathbb{S} \times \{0, 1\}$$

- Important: $\mathcal{D}_{\text{cal}}$ must be disjoint from the training data used to learn the scoring classifier to avoid bias in estimated calibration

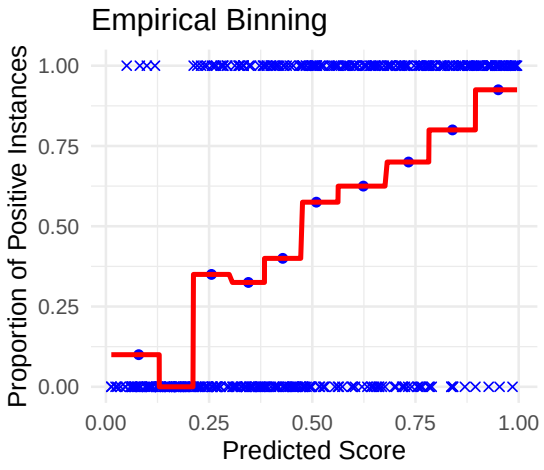
EMPIRICAL BINNING

Empirical binning partitions predicted scores $s \in \mathbb{S}$ into bins $B_1, \ldots, B_M$ and defines a (piecewise constant) calibration function C by:

$$C(s) = \bar{p}_{J(s)} = \frac{\sum_{i=1}^{n} \mathbb{1}[s^{(i)} \in B_{J(s)}] \cdot y^{(i)}}{|B_{J(s)}|}, \text{ where}$$

- $J(s) \in \{1, \ldots, M\}$ is the bin index that s belongs to (i.e., $s \in B_{J(s)}$)
- $s^{(i)} = s(\mathbf{x}^{(i)})$ is the score and $y^{(i)}$ the label of instance i
- numerator counts the number of positive instances within $B_{J(s)}$

$\Rightarrow$ C maps s to $\bar{p}_{J(s)}$ (average proportion of positive instances in $B_{J(s)}$)



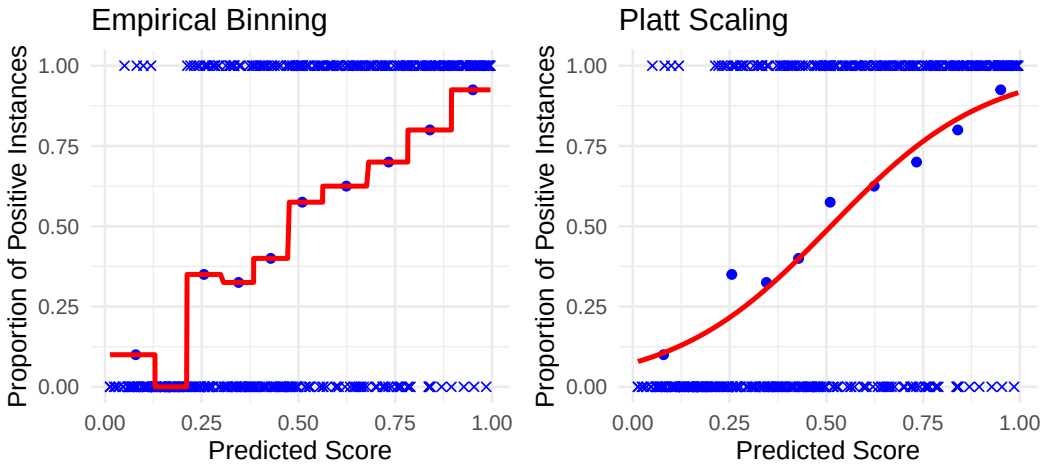
PLATT SCALING

▸ Niculescu-Mizil et al., 2005

Platt scaling applies logistic regression to predicted scores $s \in \mathbb{S}$, i.e., it fits a calibration function C minimizing the log-loss on $\mathcal{D}_{cal}$ by:

$$C(s) = \frac{1}{1 + \exp(\theta_0 + \theta_1 \cdot s)}, \quad \text{where}$$

- θ_0 is the intercept, and θ_1 the slope estimated by the logistic regression
- Requires: unbounded outputs before sigmoid/softmax, good for Neural Networks, SVM, boosted trees



ISOTONIC REGRESSION

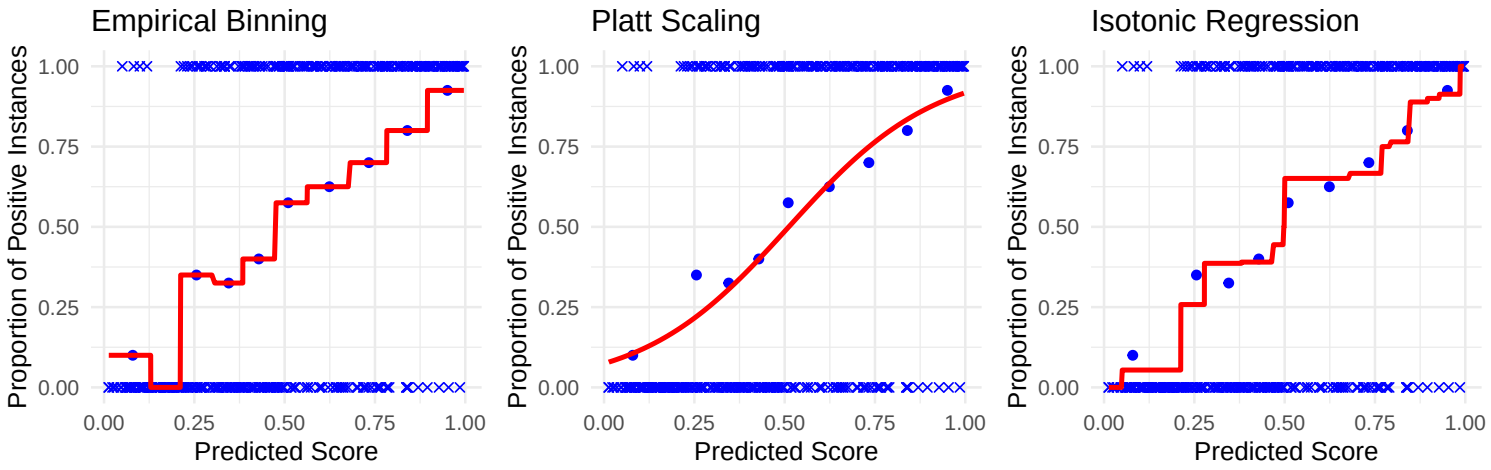
Kull et al., 2017

- *Isotonic regression* combines the non-parametric character of empirical binning with the monotonicity guarantee of Platt scaling by minimizing

$$\sum_{i=1}^{n} w_i \left(C(s^{(i)}) - y^{(i)}\right)^2$$

subject to the constraint that C is isotonic: $C(s) \leq C(t)$ for $s < t$.

- Note: C is evaluated only at a finite number of points; in-between, one may (linearly) interpolate or assume a piecewise constant function.



ISOTONIC REGRESSION - PAVA

- Optimization can be solved by the pool-adjacent violators algorithm (PAVA) by sorting scores such that

$$s^{(1)} < s^{(2)} < \ldots < s^{(n)} .$$

- Initialize bins B_i for each observation $(s^{(i)}, y^{(i)})$
- Assign to all scores $s \in B_i$: $C(s) = y^{(i)}$ with initial width $w(B_i) = 1$.
- A merge operation combines two adjacent bins B_j and B_k into a new bin $B = B_j \cup B_k$ with width $w(B) = w(B_j) + w(B_k)$ and

$$C(B) = \frac{w(B_j)C(B_j) + w(B_k)C(B_k)}{w(B_j) + w(B_k)} .$$

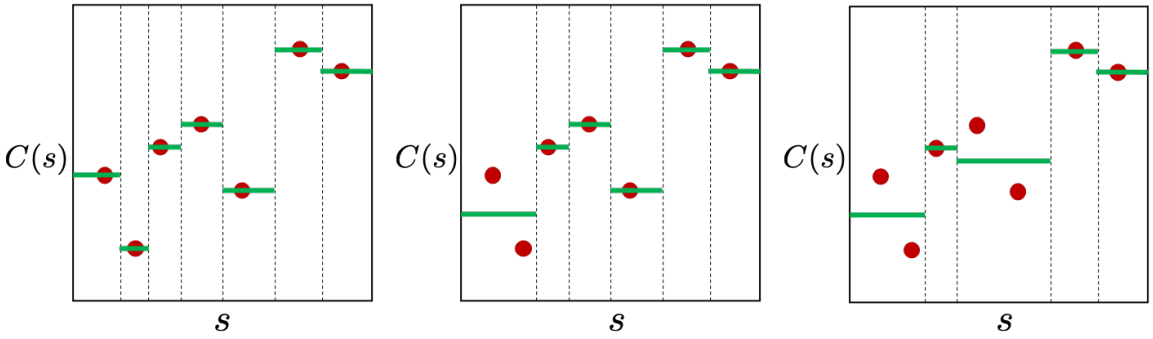
- The algorithm looks for violations of the monotonicity constraint and adjusts them with the best possible fit under the constraint ($\mathcal{O}(n)$).

ISOTONIC REGRESSION - PAVA

PAVA iterates the following steps (simplified to avoid notational overload):

(1) Find first violating adjacent bins B_i and B_{i+1} such that $C(B_i) > C(B_{i+1})$.

(2) Merge B_i and B_{i+1} into a new bin B. Stop if no violation occured.



ISOTONIC REGRESSION - PAVA

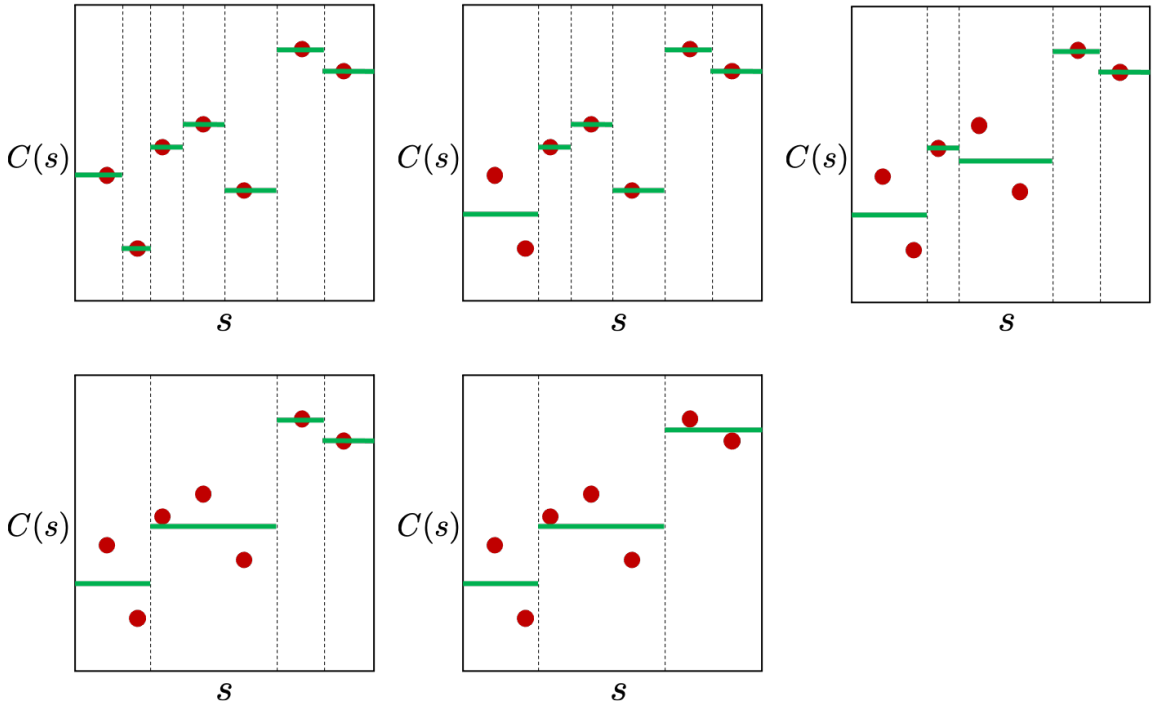
PAVA iterates the following steps (simplified to avoid notational overload):

(1) Find first violating adjacent bins B_i and B_{i+1} such that $C(B_i) > C(B_{i+1})$.

(2) Merge B_i and B_{i+1} into a new bin B. Stop if no violation occured.

(3) If $C(B) < C(B_{i-1})$ for the left neighbor bin B_{i-1}, merge also these bins and continue until no more violations are found (monotonicity).

(4) Continue with **(1)**.



HOW TO SPLIT DATA FOR CALIBRATION? ▸ Eskandar, 2023

Problem: How to avoid overfitting the calibrator during training?

Deployment scenario:

- Inducer (ML algorithm & hyperparameters) already selected
- Want to calibrate predictions of resulting model using all available data

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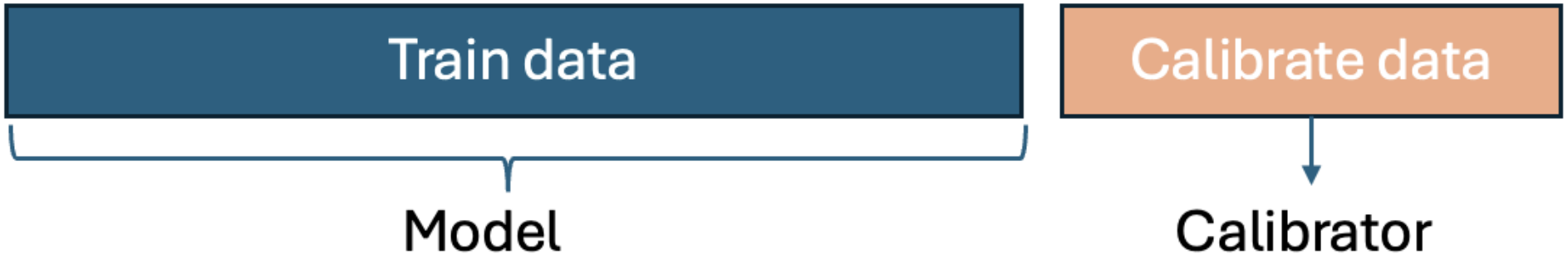
Naive solution: Use hold-out calibration set

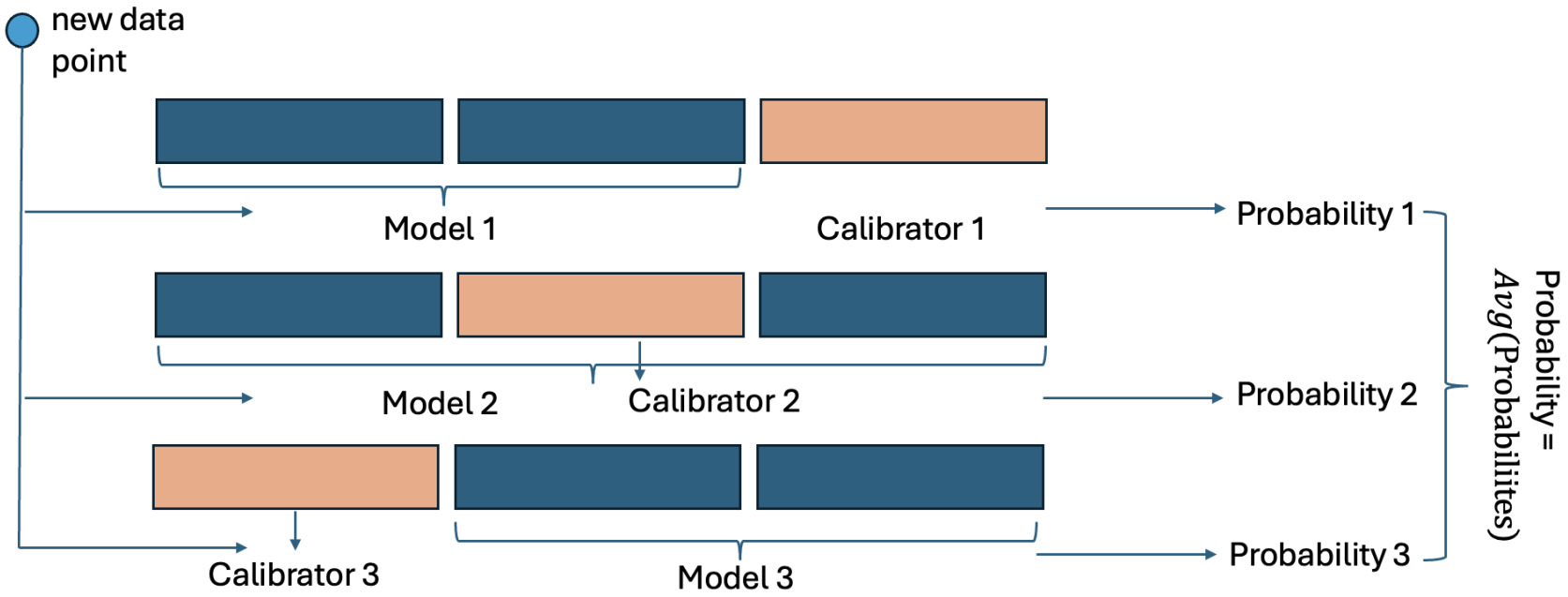


Limitation: Does not utilize full dataset for training model and calibrator

Idea: Use k-fold CV to use available data more efficiently

CV FOR CALIBRATION: PER-FOLD STRATEGY

- Fit model and calibrator on each CV fold separately
- Use fold-wise models to generate (uncalibrated) predictions and fold-wise calibrators to calibrate them
- Average fold-wise calibrated predictions



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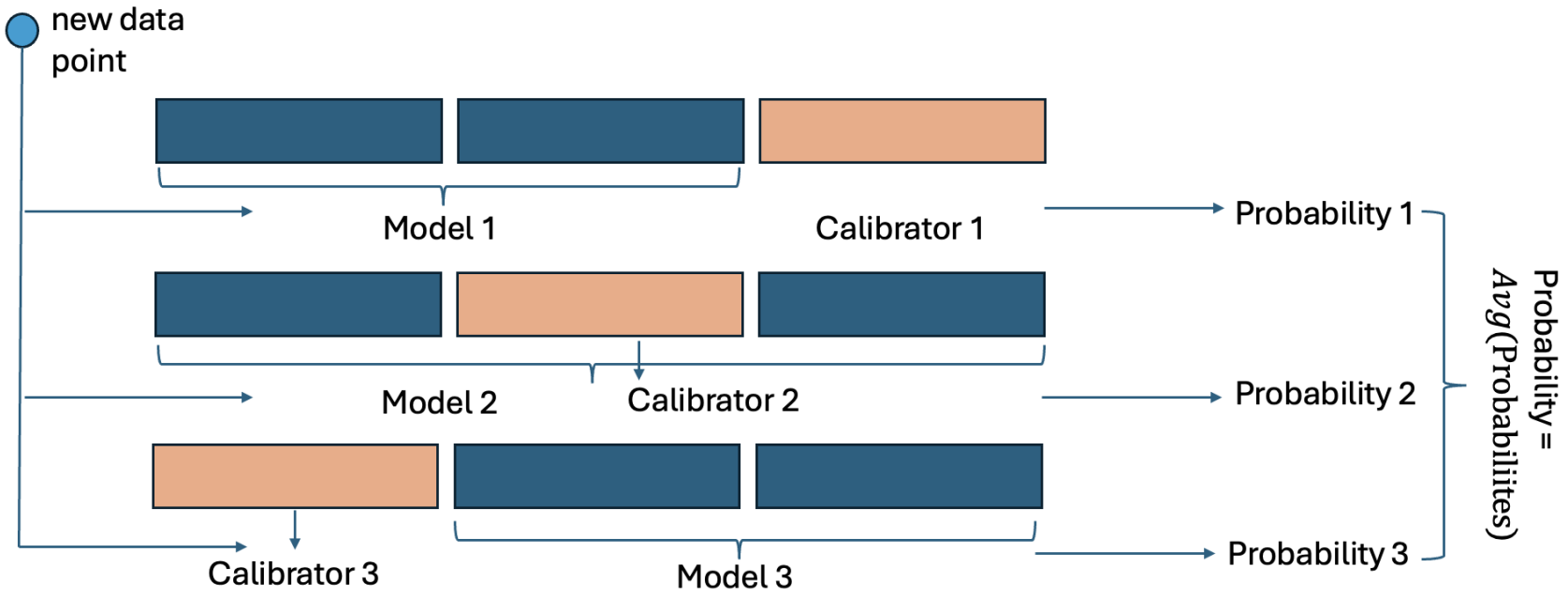
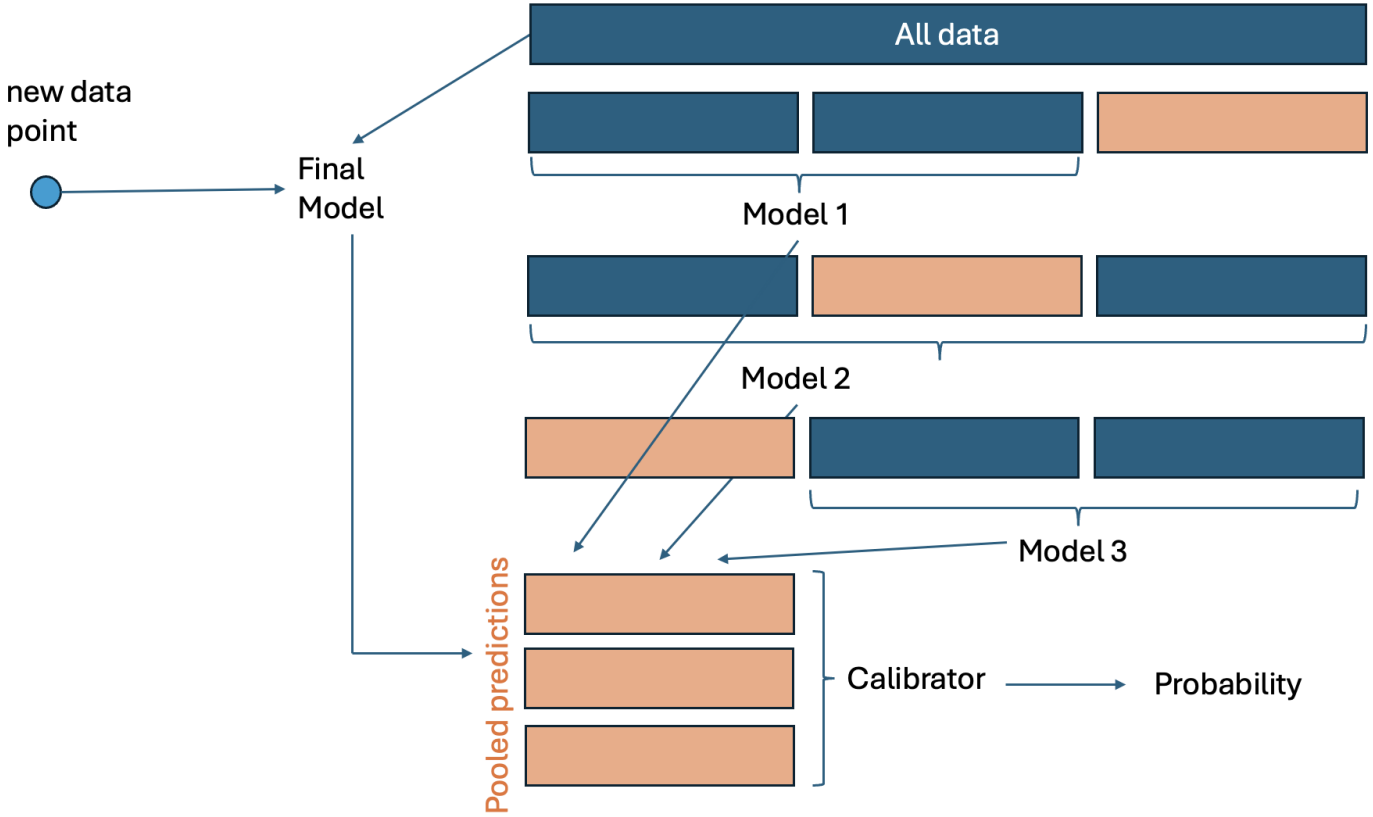


Drawback: Computationally inefficient (requires training k models and k calibrators); none are trained on full data $\Rightarrow$ pessimistic bias in calibration.
Alternative: CV with pooled out-of-fold (OOF) predictions

CV FOR CALIBRATION: POOLED OOF STRATEGY

1. Fit final model on entire dataset
2. Perform CV and generate out-of-fold predictions from CV models
3. Train calibrator on pooled OOF CV predictions (uses all observations)
4. Deploy: predict with final model, calibrate with pooled calibrator

⇒ **Advantage:** One final model + one calibrator, both trained on all data



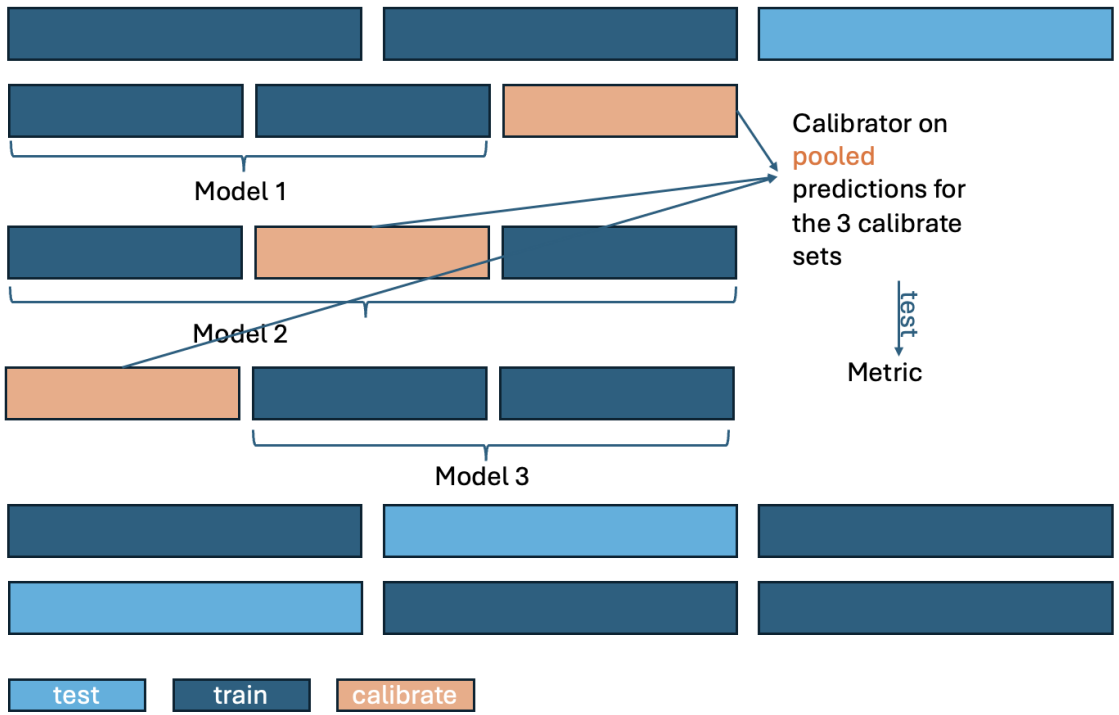
WHEN EVALUATION IS NEEDED

Setting: When the aim is to tune hyperparameters or estimate performance, we require an independent test set (besides the train and calibration set).

Option 1: Simple hold-out split (train set, calibration set, test set)

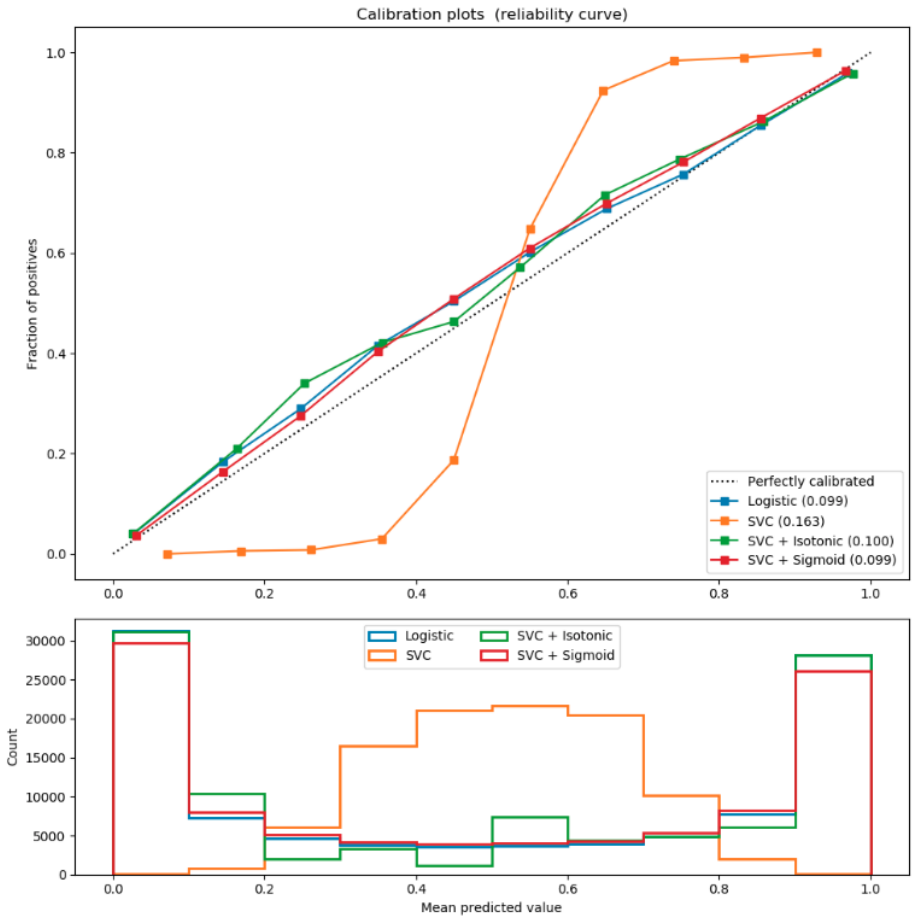
Option 2: Nested Cross-Validation

- Inner loop: Train model & calibrator (per-fold or pooled OOF strategy)
- Outer loop: Evaluate generalization performance



EXAMPLE: CALIBRATION PLOT (REVISITED)

Calibrating a SVC with Platt's scaling and isotonic regression:



$\Rightarrow$ all calibrations improved the original SVC

SUMMARY: WHY IS CALIBRATION IMPORTANT?

- **Interpreting Predicted Probabilities:**
  - Good calibration ensures that predicted probabilities accurately reflect actual risks or frequencies of events.

  $\Rightarrow$ Otherwise, predictions are just scores that happen to lie in $[0, 1]$.
- Instance-based probability evaluation metrics, such as Brier score or log-loss, always measure calibration (plus something else).
- **Example: Medical Diagnosis**
  - Poor calibration: Only half of patients with 90% predicted probability of a disease actually have it.
  - Good calibration: 90% of patients with 90% predicted probability of a disease actually have it.